

LEVELS 31-32

Chapter 29: Evaluation Rubric & Final Framework

System Design Master Roadmap — 9 Years of Experience Edition

Compiled August 2026



Chapter 29: Evaluation Rubric & The Final "Design Any Product" Framework

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29.1 The 100-Point Evaluation Rubric

Use this to self-score every mock interview from [Chapter 28](#), or have a study partner score you. Be honest — an inflated self-score is the one thing that will actually hurt you here, since the entire point is finding real gaps before a real interviewer does.

Category	Points	What earns full marks
Requirement clarification	10	Asked scoped, sharp clarifying questions (not generic ones) within the first 5 minutes; correctly identified FR vs. NFR; didn't over-ask (more than ~6 questions starts costing points for pacing)
Capacity estimation	10	Did the DAU→RPS→peak→storage math out loud, rounded sensibly, and — critically — used the numbers to justify at least one later architecture decision, rather than treating it as a disconnected ritual
Architecture	15	Clean, incremental, correctly-labeled diagram (Chapter 15's notation); logical component boundaries; sync vs. async correctly distinguished visually
Data model	10	Right database category chosen with a stated reason (Chapter 4.6's framework); key entities and relationships correctly identified; no missed obvious entity
API design	5	RESTful/appropriate method+path conventions; pagination handled if relevant; idempotency mentioned for unsafe writes where relevant
Scalability	15	Correctly identifies what needs to scale and why; uses the DB scaling ladder in the right order; doesn't over- or under-engineer relative to the stated scale
Reliability	10	Timeouts/retries/circuit breakers mentioned where genuinely relevant; replication/failover addressed; RPO/RTO discussed if disaster recovery comes up
Consistency	5	States a consistency model per data type (not one blanket answer); correctly applies CAP/PACELC reasoning where it matters
Security	5	At least 1-2 concrete, relevant security decisions raised unprompted (not generically bolted on at the very end)
Observability	5	Logging/metrics/tracing mentioned with enough specificity to show real operational thinking, not just a buzzword drop

Category	Points	What earns full marks
Trade-offs	5	Explicitly names at least 2-3 real trade-offs made during the session, unprompted
Communication	5	Thinks aloud clearly, invites the interviewer in (Chapter 14.1's Step 6 framing), handles a challenge without becoming defensive (Chapter 14.2)
Total	100	

Readiness Bands

Score	Band	What it means
0-40	Not ready	Fundamentals still shaky — go back to Chapters 1-8 before attempting more mocks; more mock volume won't help until the base concepts are solid
40-60	Beginner	Core concepts present but thin — likely missing capacity estimation discipline, trade-off articulation, or failure handling; focus revision on whichever rubric categories scored lowest, not a general re-read
60-75	Intermediate	Solid, structurally correct designs — the gap to close is usually depth on deep-dives and unprompted trade-off/security/observability mentions; this is a reasonable bar for Tier-2 product companies
75-85	Senior interview ready	Consistently strong across all categories, handles challenges well — this is your target band for the companies in Chapter 19's India/UAE list, and a real, crackable bar for many Tier-1 mid-senior loops too
85-95	Tier-1 ready	Staff-level trade-off and business-context reasoning (Chapter 14.3) showing up consistently, not just occasionally; ready for Amazon/Google/Meta-caliber senior loops
95+	Excellent	Rare in practice — treat consistently scoring here as a signal to spend remaining prep time on breadth (more unfamiliar problems) rather than more depth on problems you've already mastered

A calibration note on these thresholds: they're deliberately conservative relative to some public rubrics, reflecting that real interviewers weight unprompted trade-off articulation and failure-handling more heavily than raw technical correctness once you're past the "does this design basically work" bar — which matches everything in [Chapter 14](#) about what actually separates candidates in practice.

29.2 The Final "Design Any Product" Framework

When an interviewer says "design X" and X is something you've never specifically studied, here is the exact sequence to run — memorize this page, not the 30 worked problems.

ARCHITECTURE DIAGRAM (MERMAID SOURCE — RENDERS VISUALLY IN THE COMPANION .MD FILE / ANY MERMAID-AWARE VIEWER)

```
flowchart TD
    A[Interviewer says: Design X] --> B[1. Clarify: what does X actually do, for whom, at what scale]
    B --> C[2. Identify the users and core workflows]
    C --> D[3. Estimate scale: DAU, RPS, peak, storage]
    D --> E[4. Define 4-6 core APIs]
    E --> F[5. Define the data model and pick DB category]
    F --> G[6. Draw high-level architecture, narrating as you build it]
    G --> H[7. Identify likely bottlenecks given the numbers from step 3]
    H --> I{Does a bottleneck need a cache?}
    I -->|Yes| I1[Add cache-aside, state TTL + eviction]
    I -->|No| J
    I1 --> J{Does anything need decoupling from the request path?}
    J -->|Yes| J1[Add a queue, state what it decouples and why]
    J -->|No| K
    J1 --> K{Will one DB instance hit a ceiling?}
    K -->|Yes, evidenced by step 3's numbers| K1[Read replica to shard to distributed DB, in that order]
    K -->|No| L
    K1 --> L[8. State the consistency model per data type]
    L --> M[9. Add reliability: timeouts, retries, circuit breakers, replication]
    M --> N[10. Weave in security: auth, encryption, rate limiting]
    N --> O[11. Weave in observability: logs, metrics, traces]
    O --> P[12. State 2-3 explicit trade-offs made along the way]
    P --> Q[13. Note future scaling direction if time remains]
    Q --> R[Done – invite challenge questions and defend calmly]
```

The one-sentence version to internalize: *clarify → identify users and workflows → estimate → API → data model → architecture (narrated) → bottlenecks → cache/queue/ scale as evidenced by the numbers → consistency per data type → reliability → security → observability → trade-offs → future scaling.* This is the same 10-step framework from [Chapter 14.1](#), expanded with the two workflow-identification steps at the front that matter most for a genuinely unfamiliar product.

Why "identify users and workflows" deserves its own explicit step for unfamiliar products specifically: for a well-known product (Twitter, Uber), the core workflows are obvious and you can move straight to estimation. For something you've genuinely never considered — "design a system for matching blood donors to hospitals in real time," say — spending 2 focused minutes explicitly listing *who* uses this (donors, hospitals, coordinators) and *what they each need to do* is what lets you derive the rest of the design logically instead of freezing. This is the literal mechanism behind this roadmap's stated philosophy: **understand → visualize → implement mentally → design → explain → defend → optimize**, applied to a problem you're seeing for the first time, in real time, under pressure — which is the actual skill every other chapter in this roadmap has been building toward.

29.3 Final Readiness Checklist

Before a real interview, confirm every box below is genuinely, honestly checked — not "I read the chapter once."

- ☐ I can run the 10-step framework on a cold, unfamiliar prompt, alone, in 45 minutes, without notes.
- ☐ I can do back-of-envelope capacity estimation for any product in under 5 minutes.
- ☐ I can draw all four diagram flow types cleanly and quickly by hand.
- ☐ I can explain CAP/PACELC, consistent hashing, idempotency, and the Saga pattern from memory, precisely.
- ☐ I can explain double-entry ledger accounting and walk through a P2P transfer at the data level.
- ☐ I have scored 75+ on at least 3 mocks from the Senior tier ([Chapter 28](#)).
- ☐ I have scored 75+ on at least 2 mocks from the Tier-1 tier.
- ☐ I can name 2–3 specific, reported interview patterns for each of my top 3 target companies ([Chapter 19](#)).
- ☐ I can handle a direct challenge to my design without becoming defensive, using the validate-then-justify-the-exception pattern ([Chapter 14.2](#)).
- ☐ I've reviewed the [cheat sheet](#) within the last 3 days.
- ☐ I know my 3 weakest topics right now, and I've deliberately practiced them in the last week, not avoided them.

When every box is checked, honestly, you are ready. Go get it.

This is the final chapter of the roadmap's core content. Return to the [Index](#) for the full table of contents, or revisit the [Cheat Sheet](#) for pre-interview revision.